Example 9.3

Evaluate $\lim_{x\to 2^-}\lfloor x\rfloor$ and $\lim_{x\to 2^+}\lfloor x\rfloor$.

Solution

The greatest integer function $f(x)=\lfloor x\rfloor$ is defined as the greatest integer lesser than or equal to x.

From the graph (Fig. 1.25) of this function it is clear that $\lim_{x\to 2^-}\lfloor x\rfloor=1$ and $\lim_{x\to 2^+}\lfloor x\rfloor=2$.

Moreover, for any integer n, $\lim_{x\to n^-}\lfloor x\rfloor=n-1$ and $\lim_{x\to n^+}\lfloor x\rfloor=n$.

Does $f(x)=\lim_{x\to n}\lceil x\rceil$ exist? Look at the graph of $\lceil x\rceil$ (Fig.1.26) for the answer.

Example 9.4

Let $f(x)=\begin{cases}x+1, & x>0\\ x-1, & x<0\end{cases}$.

Verify the existence of limit as $x\to 0$.

Solution

The function is graphed in Fig.9.12.

Clearly $\lim_{x\to 0^-} f(x)=-1$ and $\lim_{x\to 0^+} f(x)=1$. Since these limits are different, $\lim_{x\to 0} f(x)$ does not exist.

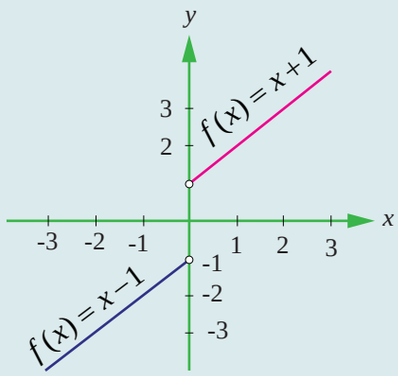


Fig. 9.12

Example 9.5

Check if $\lim_{x\to -5} f(x)$ exists or not, where $f(x)=\begin{cases}\dfrac{|x+5|}{x+5}, & \text{for } x\neq -5\\ 0, & \text{for } x=-5\end{cases}$

Solution

(i) $f(-5^-)$.

For $x<-5$, $|x+5|=-(x+5)$

Thus $f(-5^-)=\lim_{x\to -5^-}\dfrac{-(x+5)}{(x+5)}=-1$

(ii) $f(-5^+)$.

For $x>-5$, $|x+5|=(x+5)$

Thus $f(-5^+)=\lim_{x\to -5^+}\dfrac{(x+5)}{(x+5)}=1$

Note that $f(-5^-)\neq f(-5^+)$. Hence the limit does not exist.

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